

Expressions using Letter-Numbers

CHAPTER

04

Learning Outcomes...

- The Notion of Letter-Numbers
- Algebraic Expressions
- Revisiting Arithmetic Expressions
- Omission of the Multiplication Symbol in Algebraic Expressions
- Representing Situations with Algebraic Expressions
- Simplification of Algebraic Expressions
- Pick Patterns and Reveal Relationships
- Algebraic Expressions to Describe Patterns
- Patterns in a Calendar
- Matchsticks Patterns

Introduction

Expressions help us write statements using numbers and letters. Instead of writing long sentences, we use letters to represent unknown values, like the cost of one pen or the age of a person. In daily life, expressions help us calculate things such as total price, distance, or time when some information is missing. They make calculations shorter, easier, and more general for any situation. In this chapter, we will learn how to form, simplify, and use such expressions in real-life problems.

The Notion of Letter-Numbers

Letter-numbers are letters used to represent numbers whose values may vary. They help us write mathematical relations in a short and simple way. When we express relations like “Nakul’s age is 5 more than ankit’s age” using letters—for example, $a = n + 5$ —the statement becomes general and can be used for any value of n .

Algebraic Expressions

An algebraic expression is a mathematical phrase that contains letter-numbers, numbers, and operations. Expressions such as $n + 3$ show how one quantity depends on another. Algebraic expressions make it easier to describe patterns, solve problems, and understand relationships in mathematics.

Factors and Coefficients of a Term

Factors

A term of an algebraic expression can be written as a product of numbers and letter numbers. These numbers and letter numbers are known as factors of the term. Factors containing letter numbers are called algebraic factors.

For example ; In the algebraic expression, term $24x$ can be factorised as $2 \times 2 \times 2 \times 3 \times x$

Coefficients

Coefficient is usually the numerical factor of a term. A coefficient may be an algebraic factor of product of remaining factors.

For example : In the algebraic expression, $24ay$, 24 is coefficient of term ay and $24a$ is the coefficient of y .

Revisiting Arithmetic Expressions

An arithmetic expression is a combination of numbers and operations like $+$, $-$, $\times$, and $\div$.

Sometimes expressions can look confusing, so we use simple rules to make calculations easier.

Swapping (Commutative Property): We can add numbers in any order.

For example; $4 + 7 = 7 + 4$

Grouping (Associative Property): We can group numbers in any way that makes adding easier.

For example; $(8 + 13) + 5 = 8 + (13 + 5)$

Note: Swapping and grouping do not change the value of an expression.

Brackets: Brackets tell us which part to solve first.

For example; $20 + (8 \times 3) = 20 + 24$

Brackets with a Negative Sign: If a negative sign is outside the bracket, we change the signs of the numbers inside.

For example; $-(6 + 4) = -6 - 4$

Distributive Property: Multiply a number with each term inside the bracket: $a \times (b + c)$
 $= a \times b + a \times c$

For example; $5 \times (6 + 2) = 5 \times 6 + 5 \times 2$

Omission of the Multiplication Symbol in Algebraic Expressions

In arithmetic, we often use the multiplication symbol ($\times$) to show that two numbers are being multiplied. But in algebra, we have a shorter way to write multiplication especially when multiplying numbers with letters.

For example ; $7 \times k$ is written as $7k$, similarly $6 \times a + b$ is written as $6a + b$. We do not write the multiplication symbol. We simply place the number before the letter.

Representing Situations with Algebraic Expressions

In everyday life, we come across situations that involve numbers—like buying items, building patterns, or measuring distances. Instead of solving each problem again and again, we can use algebra to create short and smart expressions that work for all similar situations.

Following are some situations that can be represented by algebraic expressions:

- Real-world problems
- Repeating patterns
- Costs and quantities
- Age differences and more

Whenever something is repeated, and each unit has a fixed value, we can use the formula:

Total = (value per item) $\times$ (number of items). We use a letter (like n , t , or s) to show that the number of items can change.

For example ; Triangular flags are fixed in a row along a string. Each triangular flag in a row is fixed using two pins.

Let's observe how the total number of pins used increases as more flags are added.

Number of flags	Total pins
1	2
2	4
3	6
4	8

The numbers 2, 4, 6, 8... increase by 2 each time. This tells us the pattern is growing by equal steps.

If 1 flag needs 2 pins, then the total pins needed = $2 \times$ Number of flags.

Let's say the number of flags is f . Then the algebraic expression for total number of pins is given by $2f$.

ILLUSTRATIONS

1 There are 5 stickers in each packet, so write an expression representing number of stickers in n packets.

Soln.: Number of stickers in one packet = 5
Number of stickers in n packets = $5n$

2 One square table has 2 chairs. Every time we join another table, we can add 2 more chairs (because sides are shared). The following table shows the number of tables and chairs that can be added.

Number of tables (t)	Total chairs
1	2
2	4
3	6
4	8

How many chairs will be there if there are t tables?

Soln.: The pattern increases by 2 chairs per table.

If t is the number of tables, then the total number of chairs is $2t$.

Algebraic Expressions in Real Life

When we go for shopping, the total bill depends on how many items we buy and their unit prices. Instead of adding everything manually each time, we can create a simple algebraic expression that calculates the total cost for us – quickly and correctly!

By using letters to represent quantities, we can write general expressions that help in any situation, whether you're buying fruits, stationery, or clothes.

For example ; Ria is selling tickets for a school magic show. Each child's ticket costs ₹ 30 and each adult's ticket costs ₹ 50. She wants to calculate the total money collected based on the number of children and adult tickets sold.

Let's us create a algebraic expression that represent the money collected by Ria for different number of tickets sold.

Let c = number of child tickets sold and a = number of adult tickets sold

Now, each child ticket costs ₹ 30.

So, the total cost for c tickets = $30 \times c = 30c$

Each adult ticket costs ₹ 50.

So, the total cost for ' a ' tickets = $50 \times a = 50a$

Total amount collected = $30c + 50a$

This expression helps Ria to calculate the total amount collected for any number of children and adult tickets – just by replacing c and a with actual numbers.

ILLUSTRATIONS

3 In a school lab a black and white page costs ₹ 2 to print and a colour page costs ₹ 5 to print. If b represents number of black and white pages and c represents number of colour pages. Then, write the algebraic expression that represents the total cost.

Soln.: Cost to print 1 black and white page = ₹ 2

Cost to print b black and white pages = ₹ $2b$

Cost to print 1 colour page = ₹ 5

Cost to print c colour pages = ₹ $5c$

Hence, total cost = ₹ $(2b + 5c)$.

4 Priya bought two types of candies. Cost of candy I and II is ₹ x each and ₹ y each respectively. Write the algebraic expression to find the total cost of 2 candies of type I and 5 candies of type II.

Soln.: Cost of 1 candy of type I = ₹ x

∴ Cost of 2 candies of type I = ₹ $x \times 2 = ₹ 2x$

Cost of 1 candy of type II = ₹ y

∴ Cost of 5 candies of type II = $5 \times ₹ y = ₹ 5y$

Hence, total cost = ₹ $(2x + 5y)$.

Using Algebraic Expressions in Formulae and Rules

In mathematics, any rule, formula or pattern can be written using algebraic expressions.

Formulae for Perimeter

(i) Perimeter of an equilateral triangle = $3 \times$ length of its side

If the length of a side of an equilateral triangle is l , then its perimeter = $3l$.

(ii) Perimeter of a square = $4 \times$ length of its side

If the length of a side of square is a , then its perimeter = $4a$.

(iii) Perimeter of rectangle = $2 \times (\text{length} + \text{breadth})$.

If the length of a rectangle is l and its breadth is b , then its perimeter = $2(l + b)$.

ILLUSTRATIONS

5 A square shaped garden has all sides equal and its fencing costs ₹ 12 per metre. Write an algebraic expression that represent total cost of fencing.

Soln.: Let the length of one side be s metres.
The cost per metre is ₹ 12.

Perimeter of squares shaped garden = $4s$

∴ Expression for finding total cost
= $4s \times ₹ 12 = ₹ 48s$

Extending Arithmetic Rules to Algebraic Expressions

We learned how to simplify arithmetic expressions using rules like:

- Changing the order (Commutative property)
- Grouping terms (Associative property) using brackets and signs
- Multiplying over brackets (Distributive property).

Now, we'll see that these same properties also work for algebraic expressions, which include letter-numbers (variables).

Property 1: Commutative Property of Addition

Changing the order of terms does not change the sum.

In arithmetic: $5 + 8 = 8 + 5$

In algebra: $x + y = y + x$, for any letter-numbers (variables) x and y .

It means that the expression $2x + 3y$ can be written as $3y + 2x$. i.e., swapping (adding two terms in any order) does not change the value.

Property 2: Associative Property of Addition

Grouping of terms can be changed without affecting the result.

In arithmetic: $(4 + 6) + 5 = 4 + (6 + 5)$

In algebra: $(x + y) + z = x + (y + z)$, for any letter-numbers (variables) x , y and z .

Property 3: Distributive Property of Multiplication Over Addition

A number multiplied with a sum gives the same result as multiplying each term separately and then adding.

In arithmetic: $3 \times (6 + 2) = 3 \times 8 = 24$ and also $3 \times (6 + 2) = 3 \times 6 + 3 \times 2 = 18 + 6 = 24$

In algebra: $m(x + y) = mx + my$, where x , y are letter-numbers (variables)

This is very useful in simplifying expressions with brackets.

Property 4: Using Brackets and Signs

Brackets help in grouping terms and operations. A negative sign outside the bracket affects every term inside.

In arithmetic: $10 - (3 + 2) = 10 - 3 - 2 = 10 - 5 = 5$

In algebra: $x - (a + b) = x - a - b$, for any variables x , a and b .

ILLUSTRATIONS

6 Simplify the expression: $4(x + 6)$ and find its value for $x = 12$.

Soln.: $4(x + 6) = 4 \times x + 4 \times 6 = 4x + 24$

Now, for $x = 12$, $4x + 24 = 4 \times 12 + 24$
 $= 48 + 24 = 72$

7 Simplify the expression $9a - (3a + 4)$ and find its value for $a = 5$.

Soln.: $9a - (3a + 4) = 9a - 3a - 4$

Now, for $a = 5$, $9a - 3a - 4$
 $= 9 \times 5 - 3 \times 5 - 4 = 45 - 15 - 4 = 26$

Simplification of Algebraic Expressions

Just like we simplify long arithmetic expressions by combining like numbers, we can simplify algebraic expressions by combining like terms. This makes expressions shorter, easier to read, and faster to evaluate. Firstly, let us learn about like and unlike terms in an algebraic expression.

Like Terms

In an algebraic expression, the terms having the same letter-numbers are known as like terms.

Unlike Terms

Unlike terms do not have the same letter-numbers.

For example : In the expression $8x - 2y - x$, $8x$ and $-x$ are like terms, where $8x$ and $-2y$ are unlike terms.

Addition and subtraction in an algebraic expression can be performed only on like terms.

In expression $3y - 2y + 2x$, like terms are $3y$ and $-2y$, this expression can be further solved as,

$$3y - 2y + 2x = (3 - 2)y + 2x = 1y + 2x = y + 2x$$

Note : Unlike terms in an algebraic expression cannot be added or subtracted.

ILLUSTRATIONS

8 Combine the numbers shown in picture. Try grouping the numbers in different ways to check if you still get the same result.



Soln.: Adding the numbers row - wise, we get

$$[-2 + (-6) + 5k + (-5g)] + [3 + 2 + 5k + 5k]$$

$$= [-8 + 5k + (-5g)] + [5 + 10k]$$

$$= -8 + 5 + 5k + 10k - 5g = 15k - 5g - 3$$

Adding like terms together gives, we get

$$-2 + (-6) + 3 + 2 + 5k + 5k + 5k - 5g$$

$$= -3 + 15k - 5g$$

Adding the numbers column-wise, we get

$$[-2 + 3] + [-6 + 2] + [5k + 5k] + [-5g + 5k]$$

$$= 1 - 4 + 10k - 5g + 5k = -3 + 15k - 5g$$

Yes, we get the same result by adding terms in any way.

9 Simplify : $3x - (-4y) + 3y - 4x - (-5x) - 6y$

Soln.: $3x - (-4y) + 3y - 4x - (-5x) - 6y$

$$= 3x + 4y + 3y - 4x + 5x - 6y$$

$$= 3x - 4x + 5x + 4y + 3y - 6y$$

$$= (3 - 4 + 5)x + (4 + 3 - 6)y = 4x + y$$

10 Subtract $3x + 4y - 2z$ from the sum of $3 + 4z + 3y - x$ and $3x + 2y - 3z + 2$.

Soln.: Sum of $3 + 4z + 3y - x$ and $3x + 2y - 3z + 2$

$$= (-x + 3y + 4z + 3) + (3x + 2y - 3z + 2)$$

$$= (3 - 1)x + (3 + 2)y + (4 - 3)z + (3 + 2)$$

$$= 2x + 5y + z + 5$$

Now, required difference

$$= (2x + 5y + z + 5) - (3x + 4y - 2z)$$

$$= (2 - 3)x + (5 - 4)y + (1 + 2)z + 5$$

$$= -x + y + 3z + 5$$

11 Each row in a classroom has 6 chairs. There are r rows in the front and $(r + 2)$ rows in the back. Find the expression that represents total number of chairs in the classroom.

Soln.: Number of chairs in front rows = $6r$

Number of chairs in back rows = $6(r + 2)$

$$= 6r + 12$$

Total number of chairs in classroom

$$= 6r + 6r + 12 = 12r + 12$$

Mind the Mistake, Mend the Mistake

Sometimes, while simplifying algebraic expressions or replacing letter-numbers with numbers, mistakes happen. So, we need to learn to spot the error, understand what went wrong, and correct it properly.

Common Mistakes

- Incorrect sign handling: The negative sign $(-a)$ was wrongly taken as positive $(+a)$.
- Partial distribution: $5(2a + 3) \neq 10a + 3$, Multiply every term inside the bracket with outside constant.
- Incorrect multiplication of negatives: $-2(x + 5) \neq -2x + 10$, Here $-2 \times 5 = -10$ as, positive $\times$ negative = negative

- **Wrong removal of brackets with minus sign:** When there is a minus sign before a bracket, change signs of all terms inside the bracket.
- **Combining unlike terms:** Only like terms can be added or subtracted with each other.

For example; If we have an expression say $2a - 3b$ and value of $a = 2$ and $b = -3$, then we need to take care of negative sign while replacing the values of a and b .

Since $b = -3$, so value of $2a - 3b = 2 \times 2 - (3 \times (-3)) = 4 + 9 = 13$

Common mistake made in such examples is that we do not take care of negative sign outside bracket and write it as $2a - 3b = 2 \times 2 - (3 \times (-3)) = 4 - 9 = -5$, which gives us incorrect value.

ILLUSTRATIONS

12 Steps to simplify the expression $2x + 3 - 4(x - 2)$ are given. However one of the steps is incorrect. Identify the incorrect step and solve the expression correctly

Step 1 : $2x + 3 - 4(x - 2)$

Step 2 : $2x + 3 - 4x + 8$

Step 3 : $2x - 4x + 3 + 8$

Step 4 : $2x + 11$

Soln.: Here, step 4 is incorrect as correct step is $-2x + 11$ as $2x - 4x = -2x$.

Pick Patterns and Reveal Relationships

Patterns often based on some rules. In this topic, we'll learn about spotting patterns, uncovering the hidden operations behind them, and expressing those operations using simple formulae.

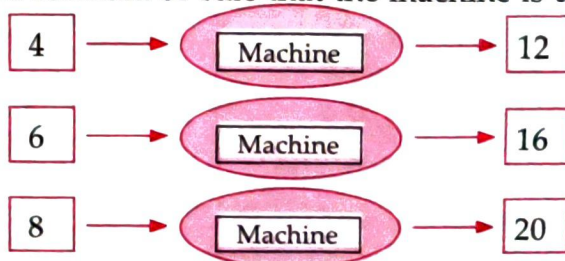
We'll explore number machines, repeating designs, and relatable situations. Each one follows a rule, and we'll learn how to figure it out. Sometimes, we'll analyse and simplify a pattern; other times, we'll create our own pattern.

By observing how inputs relate to outputs, we'll learn to:

- Detect consistent patterns across examples
- Represent them with clear and logical expressions
- Check whether our rule works in all cases.

Note that after expressing general formula for a pattern its better to check it twice so that we can omit the possibility of mistakes.

Let's start with an example, here a machine in which we are inserting numbers as input and based on some operations it provides an output, we need to observe the input and output in all cases and need to find the formula or rule that the machine is using to generate the output.



Here, we have given three cases, let's calculate the difference between input and output :

- For input 4, Output = 12 and $12 - 4 = 8$
- For input 6, Output = 16 and $16 - 6 = 10$
- For input 8, Output = 20 and $20 - 8 = 12$

Now, look at the pattern of change:

- Input increases by 2 ($4 \rightarrow 6 \rightarrow 8$)
- Output increases by 4 ($12 \rightarrow 16 \rightarrow 20$)

Let's see what happens when we multiply the input by 2:

- $2 \times 4 = 8$, then add 4 $\rightarrow 8 + 4 = 12$
- $2 \times 6 = 12$, then add 4 $\rightarrow 12 + 4 = 16$
- $2 \times 8 = 16$, then add 4 $\rightarrow 16 + 4 = 20$

It works for all the numbers. We are multiplying the input number by 2 and then adding 4. So, the rule is $\text{Output} = 2 \times \text{Input} + 4$.

We can write this using a letter like x for the input and $\text{output} = 2x + 4$.

Hence, the rule that machine uses : "Double the input and add 4" and the algebraic expression is $2x + 4$.

ILLUSTRATIONS

13 Each machine takes two inputs and performs the same hidden rule to produce the output. Your job is to find the rule the machine is using, write the rule using an algebraic expression and check your rule with all input sets. Also find the output for fourth row by using the same rule.

Input A	Input B	Output
3	5	24
6	2	30
4	7	40
7	1	?

Soln.: Let's try to guess what the machine might be doing with the two numbers.

We'll test a few ideas using the first row:

Input $A = 3$, input $B = 5$ and output = 24

Let's try multiplying the numbers: $3 \times 5 = 15$, but not 24.

Now, try adding the inputs: $3 + 5 = 8$, Still not 24.

Now, $3 \times 5 = 15$, and $15 + (3 \times 3) = 15 + 9 = 24$

Try the same with the second row: $A = 6$, $B = 2$

Now, $6 \times 2 = 12$ and $3 \times A = 3 \times 6 = 18$ and $12 + 18 = 30$

For third row: $A = 4$, $B = 7$

Now, $4 \times 7 = 28$ and $3 \times A = 3 \times 4 = 12$ and

$$28 + 12 = 40$$

So, the machine is doing, multiply input A and input B , then add 3 times input A .

For fourth row, $7 \times 1 = 7$ and $3 \times 7 = 21$ so, $7 + 21 = 29$.

Hence output for fourth row is 29.

14 A machine gives an output using the formula $3A + 2B$. Complete the following table.

Input A	Input B	Output $3A + 2B$
2	4	
3	1	
5	2	
4	3	
1	6	

Soln.:

Input A	Input B	Output $3A + 2B$
2	4	$3 \times 2 + 2 \times 4 = 6 + 8 = 14$
3	1	$3 \times 3 + 2 \times 1 = 9 + 2 = 11$
5	2	$3 \times 5 + 2 \times 2 = 15 + 4 = 19$
4	3	$3 \times 4 + 2 \times 3 = 12 + 6 = 18$
1	6	$3 \times 1 + 2 \times 6 = 3 + 12 = 15$

Algebraic Expressions to Describe Patterns

Have you ever seen a beautiful design repeating along the border of a saree, a wall, or a floor, and patterns made by matchsticks? These repeating patterns follow regular positions and algebraic expressions help us to describe exactly what each design shows up. For defining a algebraic expression for a given pattern, we follow the given steps:

Observe the pattern carefully : Look at the numbers, objects, or positions of objects in the pattern and try to find what repeats in the pattern or how it changes.

Identify the rule or relationship : Observe that how does one term change to the next? Is it increasing by a fixed number or not? Is something added or subtracted?

Find the position-based change : See how the pattern is related to its position (1st, 2nd, 3rd term, etc.). Write the output for each position.

Write the general rule using a letter (like n) : Use a letter-number (like n) to represents the position. Then, write a formula or an algebraic expression based on the rule. For example ; If every term is 3 times the position, write the expression as $3n$.

Test your expression : Try your expression for different values of n (like 1, 2, 3...) and check if it gives the correct terms in the pattern or not.

Refine if needed : If your expression doesn't match the pattern, recheck your rule and make changes.

For example ; Let us consider books are placed in a pattern on each shelf. Each shelf has 5 more books than the previous one and the first shelf has 8 books. Can you determine the number of books on 8th shelf?

Here is a pattern where each term increases by a fixed number.

Let's list the first few shelves:

1st shelf: 8 books ; 2nd shelf: $8 + 5 = 13$ books ; 3rd shelf: $13 + 5 = 18$ books ; 4th shelf: $18 + 5 = 23$ books ... and so on.

We see that it's increasing by 5 every time. So, instead of adding again and again, we can write a formula:

Let's use a letter n for the shelf number.

Shelf 1: Number of books = $8 + 0 \times 5$; Shelf 2: Number of books = $8 + 1 \times 5$; Shelf 3: Number of books = $8 + 2 \times 5$, Shelf n : Number of books = $8 + (n - 1) \times 5$

So, number of books on 8th shelf = $8 + (8 - 1) \times 5 = 8 + 7 \times 5 = 8 + 35 = 43$.

So, an algebraic expression can be formed for repeated patterns.

Patterns in a Calendar

Grids, like those seen in calendars, number tables, or tiles arrangement, often contain interesting patterns. These patterns repeat in a certain way, row by row, column by column, or diagonally. While we can observe a few of these patterns by looking into it. But for some, we need to form an algebraic expressions to understand the patterns.

For example ; Look at a calendar. Pick any 2×2 square of dates. Let's see what happens if we observe the diagonal pairs:

Take 2×2 square as shown below from this calendar:

15	16
22	23

Let's add them diagonally:

$15 + 23 = 38$ and $16 + 22 = 38$

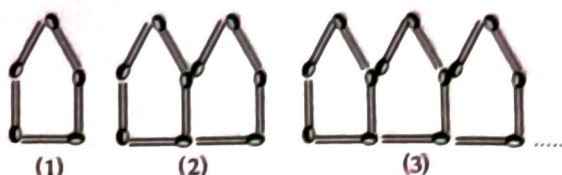
Both diagonals add up to the same number.

JULY 20XX

SUN	MON	TUE	WED	THU	FRI	SAT
		1	2	3	4	5
6	7	8	9	10	11	12
13	14	15	16	17	18	19
20	21	22	23	24	25	26
27	28	29	30	31		

ILLUSTRATIONS

17 Find the pattern followed in given figure and the number of matchsticks used in 88th figure.



Soln.:

Number of pentagons	1	2	3		88
Number of matchsticks used	5	9	13		?

From the table, a general rule is obtained.

For fig. (1), we need $4 \times 1 + 1 = 5$ matchsticks

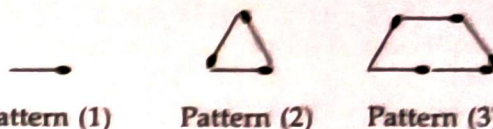
For fig. (2), we need $4 \times 2 + 1 = 9$ matchsticks

For fig. (3), we need $4 \times 3 + 1 = 13$ matchsticks

If we need ' n ' number of pentagons, then we need $4 \times n + 1 = (4n + 1)$ matchsticks.

$\therefore$ If $n = 88$, then number of matchsticks used
 $= 4 \times 88 + 1 = 352 + 1 = 353$

18 Study the pattern given below and identify the rule used in n^{th} pattern.



Soln.: For pattern 1, we have $1 \times 2 - 1 = 1$ matchstick

For pattern 2, we have $2 \times 2 - 1 = 3$ matchsticks

For pattern 3, we have $3 \times 2 - 1 = 5$ matchsticks

So, for n^{th} pattern, we have $n \times 2 - 1$

$= (2n - 1)$ matchsticks

Equation

An equation means two mathematical expressions are equal. It consists of two expressions separated by an equal sign ($=$), means both sides have the same value.

The expression on the left of the equality sign is denoted by L.H.S. and those on the right is denoted by R.H.S.

Note : If L.H.S. is not equal to R.H.S., then we do not get an equation.

COMPETITION WINDOW

How it works?

Let's say the top-left number of any 2×2 square is 'x'.

Then, the other three numbers will be:

Top-right: $x + 1$; Bottom-left: $x + 7$ and Bottom-right: $x + 8$

This is because we are assuming 7 numbers per row, like in a calendar.

Now let's add the diagonals:

First diagonal: $x + (x + 8) = 2x + 8$

Second diagonal: $(x + 1) + (x + 7) = 2x + 8$

This proves that no matter where we start, the diagonals in a 2×2 square on the calendar will always add to the same total.

There are also some other patterns also. For example ; 3rd of July 2025 is Thursday, then each 7th number from there (10, 17, 24...) also falls on a Thursday.

x	x + 1
x + 7	x + 8

ILLUSTRATIONS

- 15** Look at the calendar for August 20XX. Focus on **Wednesdays**: 7, 14, 21, 28. What pattern do you see?

August 20XX

Sun	Mon	Tue	Wed	Thu	Fri	Sat
				1	2	3
4	5	6	7	8	9	10
11	12	13	14	15	16	17
18	19	20	21	22	23	24
25	26	27	28	29	30	31

Soln.: Each number is 7 more than the previous one.

So, if the first Wednesday is w , then the next Wednesdays are: $w + 7$, $w + 14$ and $w + 21$.

- 16** Pick a plus-shaped group of 5 numbers from the calendar August 20XX as shown in

previous illustration. Add them all and find its relation with the number given in centre. Is this hold for all the grids of plus shaped?

Soln.: Let's pick a plus (+) shaped group of 5 numbers from the above calendar.

Now, let us add all the numbers.

$$7 + 13 + 14 + 15 + 21 = 70 = 5 \times 14$$

Sum of all five numbers = $5 \times$ number in the centre.

No matter what number is at the centre (say a), the numbers in the plus shape will be $a - 7$ (above), $a + 7$ (below), $a - 1$ (left and $a + 1$ (right).

Sum of these numbers

$$= (a - 7) + (a + 7) + (a - 1) + (a + 1) + a$$

$$= 5a = 5 \times \text{number in the centre.}$$

	7	
13	14	15
	21	

Matchsticks Pattern

A matchstick pattern is a sequence of shapes or figures made using matchsticks, where each new figure follows a rule or pattern. By observing how the number of sticks increases or changes from one figure to the next, we can form a general rule or formula to find the number of sticks in any figure of the pattern.

CONCEPT MAP

EXPRESSIONS USING LETTER - NUMBERS

ALGEBRAIC EXPRESSIONS

Mathematical Expressions containing letter-numbers known as algebraic expressions

Terms

A term is a part of an expression that is separated by a plus (+) or minus (-) sign.

Factors

A term of an algebraic expression can be written as a product of numbers and letter-numbers. These numbers and letter numbers are known as factors of the term.

Coefficient

Coefficient is usually the numerical factor of a term.

Like

In an algebraic expression, the terms having the same letter-numbers are known as like terms.

Unlike

Unlike terms do not have the same letter-numbers.

Simplification

Just like we simplify long arithmetic expressions by combining like numbers, we can simplify algebraic expressions by combining like terms. Addition and subtraction in an algebraic expression can be performed only on like terms.

Properties

Property 1: Commutative Property of Addition:

Changing the order of terms does not change the sum.

Property 2: Associative Property of Addition:

Grouping of terms can be changed without affecting the result.

Property 3: Distributive Property of Multiplication Over Addition:

A number multiplied with a sum gives the same result as multiplying each term separately and then adding.

Property 4: Using Brackets and Signs:

Brackets help in showing grouping terms and operations. A negative sign outside the bracket affects every term inside.

Patterns in a Calendar

Grids, like those seen in calendars, number tables, or tiles arrangement, often contain interesting patterns. These patterns repeat in a certain way, row by row, column by column, or diagonally. While we can observe a few of these patterns by looking into it. But for some we need to form an algebraic expressions to understand the patterns.

Omission of the Multiplication Symbol

In arithmetic, expression we often use the multiplication symbol ($\times$) to show that two numbers are being multiplied. But in algebra, we have a shorter way to write multiplication especially when multiplying numbers with letters.

For example: $7 \times k$ written as $7k$.

Algebraic Expressions in Formulae and Rules

Formulae for Perimeter

- (i) Perimeter of an equilateral triangle = $3 \times$ length of its side
If the length of a side of an equilateral triangle is l , then its perimeter = $3l$.
- (ii) Perimeter of a square = $4 \times$ length of its side
If the length of a side of square is a , then its perimeter = $4a$.
- (iii) Perimeter of rectangle = $2 \times$ (length + breadth).
If the length of a rectangle is l and its breadth is b , then its perimeter = $2(l + b)$.

Algebraic Expressions to Describe Patterns

For defining an algebraic expression for a given pattern, we follow the given steps:

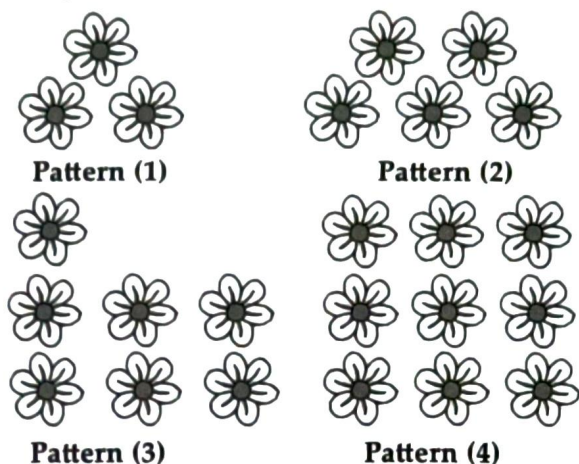
- Observe the pattern carefully
- Identify the rule or relationship
- Find the position-based change
- Write the general rule using a letter (like n)
- Test your expression
- Refine if needed

Matchsticks Pattern

A matchstick pattern is a sequence of shapes or figures made using matchsticks, where each new figure follows a rule or pattern. By observing how the number of sticks increases or changes from one figure to the next, we can form a general rule or formula to find the number of sticks in any figure of the pattern.

Solved Examples

1. Study the patterns and identify the rule for n^{th} pattern.



Soln.: For pattern 1, number of flowers
 $= 2 \times 1 + 1 = 3$

For pattern 2, number of flowers $= 2 \times 2 + 1 = 5$

For pattern 3, number of flowers $= 2 \times 3 + 1 = 7$

For pattern 4, number of flowers $= 2 \times 4 + 1 = 9$

In the same way, for n^{th} pattern, number of flowers $= 2n + 1$

2. Use the algebraic expressions to complete the following table.

S.No.	Series	Algebraic expression	Terms to be found	Term
(i)	5, 8, 11, ...	$3y + 2$	75^{th}	
(ii)	4, 6, 8, 10, ...	$2x + 2$	250^{th}	
(iii)	4, 5, 6, ...	$n + 3$	195^{th}	
(iv)	13, 21, 29, ...	$8p + 5$	68^{th}	
(v)	2, 7, 12, ...	$5z - 3$	46^{th}	

Soln.:

S. No.	Series	Algebraic expression	Terms to be found	Term
(i)	5, 8, 11, ...	$3y + 2$	75^{th}	$3 \times 75 + 2 = 227$
(ii)	4, 6, 8, 10, ...	$2x + 2$	250^{th}	$2 \times 250 + 2 = 502$
(iii)	4, 5, 6, ...	$n + 3$	195^{th}	$195 + 3 = 198$
(iv)	13, 21, 29, ...	$8p + 5$	68^{th}	$8 \times 68 + 5 = 549$
(v)	2, 7, 12, ...	$5z - 3$	46^{th}	$5 \times 46 - 3 = 227$

3. If $A = 3a - 2b$, $B = -2a + 3b$ and $C = -a - b$, then find $A + B + C$.

Soln.: $A = 3a - 2b$, $B = -2a + 3b$, $C = -a - b$

Now, $A + B + C = 3a - 2b - 2a + 3b - a - b$

$= 3a - 2a - a - 2b + 3b - b = 0$

4. A sloped rectangular roof is made of two equal slanted parts, each of length l metre and width w metres. The cost of tiling is ₹75 per square metre.

Soln.: Area of one slanted part $= 1 \times w$ sq. m

Total area 2 slanted parts $= 2 \times 1 \times w = 2w$ sq. m

Cost of tiling of 1 sq. m $= ₹75$

$\therefore$ Total cost $= ₹75 \times 2w = ₹150w$

5. Use algebraic properties to simplify the expression $2x + 3(x - 2y) + 4y - x$. Also, find the value of expression for $x = 2$ and $y = 3$.

Soln.: We have, $2x + 3(x - 2y) + 4y - x$

$= 2x + 3x - 6y + 4y - x$

[By using distributive property]

$= (2x + 3x - x) + (-6y + 4y)$ [Grouping of like terms]

$= 4x - 2y$

For $x = 2$ and $y = 3$, $4x - 2y = 4 \times 2 - 2 \times 3 = 8 - 6 = 2$

6. While simplifying the expression $x - (2y - x) + 4y$, a student writes :
 $x - 2y - x + 4y = 2y + 4y = 6y$

Was this simplification done correctly? Explain.

Soln.: Given Expression is $x - (2y - x) + 4y$

Student's attempt: $x - 2y - x + 4y = 2y + 4y = 6y$

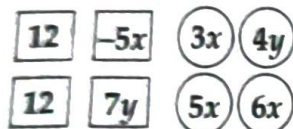
The mistake is in removing the brackets incorrectly. When we have a minus sign before brackets, we must change the sign of every term inside the bracket.

So, $x - (2y - x) = x - 2y - x$

The correct way is : $x - (2y - x) + 4y$

$= x - 2y + x + 4y = (x + x) + (4y - 2y) = 2x + 2y$

7. Combine the terms shown in figure. Try grouping the terms in different ways to check if you still get the same result. Which is the easiest and faster way of simplifying the expression?



Soln.: Adding the numbers row-wise:

$12 - 5x + 3x + 4y + [12 + 7y + 5x + 6x]$

$12 - 2x + 4y + 12 + 7y + 11x$

$12 + 12 + (-2x + 11x) + (4y + 7y) = 24 + 9x + 11y$

Adding the numbers column-wise :

$2 + 12 + [-5x + 7y] + [3x + 5x] + [4y + 6x]$

$24 - 5x + 7y + 8x + 4y + 6x$

$= 24 + (-5x + 8x + 6x) + (7y + 4y) = 24 + 9x + 11y$

Adding like terms together gives :

$(12 + 12) + (-5x + 3x + 5x + 6x) + (4y + 7y)$

$= 24 + 9x + 11y$

Although the expressions may appear different at first, they simplify to give the same final result. The fastest and the easiest way to simplifying the expression is to combining the like terms.

8. A laundry service charges ₹12 for each shirt and ₹15 for each pant. A customer gives s shirts and p pants every week for 7 weeks. Form an algebraic expression for:

(i) Total weekly cost.

(ii) Total cost over 7 weeks.

(iii) If the customer gets a discount of ₹ d each week, express the net payment over 7 weeks.

Soln.: (i) Cost for s shirts = ₹ $12 \times s = ₹12s$

Cost for p pants = ₹ $15 \times p = ₹15p$

Total weekly cost = ₹ $(12s + 15p)$

(ii) Cost for one week is ₹ $(12s + 15p)$

Total cost over 7 weeks = ₹ $7(12s + 15p)$
 $= ₹(84s + 105p)$

(iii) If the customer gets a discount of ₹ d each week

Total discount for 7 weeks = ₹ $7d$

Net payment = Total cost - Total discount

$= ₹(84s + 105p - 7d)$

9. 6 is subtracted from four times a number. Represent it algebraically and find the value at $x = 20$.

Soln.: Let the number be x .

Four times of $x = 4 \times x = 4x$

Now, 6 is subtracted from $4x$, then we have $4x - 6$

At $x = 20$, $4x - 6 = 4(20) - 6 = 80 - 6 = 74$

10. In the given calendar, if a 3×3 grid of dates is selected as shown below, if the middle date is labelled as ' x ', then write algebraic expressions for the other dates.

May 20XX

Sun	Mon	Tues	Wed	Thu	Fri	Sat
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30
31						

	x	

Soln.: The given 3×3 grid has $x = 20$

12	13	14
19	$x = 20$	21
26	27	28

So, $13 = 20 - 7 = x - 7$

$12 = 20 - 8 = x - 8$,

$14 = 20 - 6 = x - 6$

$19 = 20 - 1 = x - 1$,

$21 = 20 + 1 = x + 1$

$26 = 20 + 6 = x + 6$,

$27 = 20 + 7 = x + 7$,

$28 = 20 + 8 = x + 8$.

11. A garden bed is in the shape of a large rectangle. It is split into vegetable area of length $7a$ units and flower area of length $5a$ units. The breadth of the garden is y units. Find the total area by

- (i) adding area of both sections.
(ii) using total length of garden.

Soln.: (i) We treat each part as a smaller rectangle:

Area of vegetable section = length $\times$ breadth

$$= 7a \times y = 7ay \text{ sq. units}$$

Area of flower section = length $\times$ breadth

$$= 5a \times y = 5ay \text{ sq. units}$$



$$\text{Total area} = 7ay + 5ay = 12ay$$

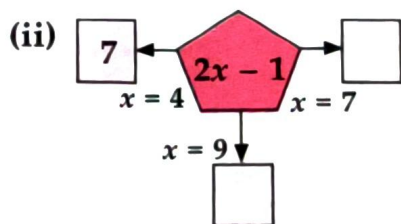
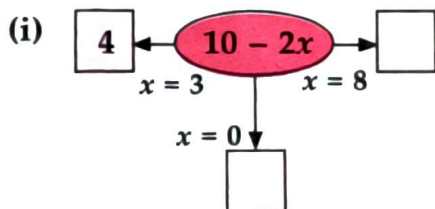
(ii) Total length of garden = $7a + 5a = 12a$ units

Breadth of garden = y units

So, total area = length $\times$ breadth = $12a \times y$

$$= 12ay \text{ sq. units}$$

12. Replace the letter-numbers with actual numbers to complete the blanks below. An example has been provided for your reference.

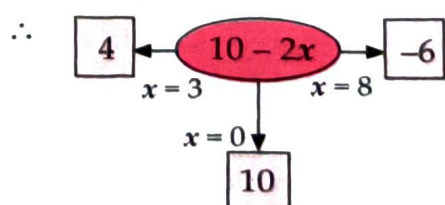


Soln.: (i) We have the expression as $10 - 2x$

for $x = 3$, $10 - 2x = 10 - 6 = 4$

For $x = 8$, $10 - 2x = 10 - 16 = -6$

For $x = 0$, $10 - 2x = 10 - 0 = 10$.

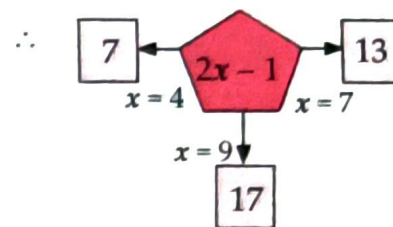


(ii) We have the expression as $2x - 1$

For $x = 4$, $2x - 1 = 8 - 1 = 7$

For $x = 7$, $2x - 1 = 14 - 1 = 13$

For $x = 9$, $2x - 1 = 18 - 1 = 17$.



13. A rectangular paper is folded once and a hole is punched through it. When unfolded, you see 2 holes. If the paper is folded twice and punched once, 4 holes appear. How many holes will there be if the paper is folded 6 times and punched once?

Can you write an algebraic expression for the number of holes if the paper is folded f times and punched once?

Soln.: Observe the given table

Number of Folds (f)	Number of Holes
1	2
2	4
3	8
4	16
5	32
6	?

We can observe that with each fold, the number of holes doubles. This is a power of 2 pattern.

So, the number of holes for f folds = 2^f

If the paper is folded 6 times, then number of holes = $2^6 = 64$

14. A tile staircase grows like this :

Step 1 : 1 tile

Step 2 : 3 tiles

Step 3 : 5 tiles

Step 4 : 7 tiles

Observe the pattern. How many tiles are needed to build n steps? Write the algebraic expression for this.

Soln.: We have,

Step 1: 1 tile = $2 \times 1 - 1$

Step 2: 3 tiles = $2 \times 2 - 1$

Step 3: 5 tiles = $2 \times 3 - 1$

Step 4: 7 tiles = $2 \times 4 - 1$

That means the pattern is an odd number sequence: 1, 3, 5, 7, 9, 11,...

So, tile count at step n is $2n - 1$.

15. In a hall, 3 new benches are added each day. On the first day, there were 5 benches. Write an expression for the total number of benches after d days.

Soln.: We have given that on Day 1, there are 5 benches and, on each day, 3 new benches are

added. So, we have,

Day 1 5 benches

Day 2 $5 + 3 = 8$ benches

Day 3 $5 + 3 + 3 = 11$ benches

Day 4 $5 + 3 + 3 + 3 = 14$ benches and so on.

We can see that after day 1, we are adding $3(d - 1)$ to the original 5 benches (because no benches are added on the first day).

Total benches after d days = $5 + 3(d - 1)$ or $3d + 2$

NCERT Section

Figure it Out

(Page 84 & 85)

1. Write formulas for the perimeter of:

(a) triangle with all sides equal.

(b) a regular pentagon (as we have learnt last year, we use the word 'regular' to say that all sidelengths and angle measures are equal)

(c) a regular hexagon

2. Munirathna has a 20 m long pipe. However, he wants a longer watering pipe for his garden. He joins another pipe of some length to this one. Give the expression for the combined length of the pipe. Use the letter-number 'k' to denote the length in meters of the other pipe.

3. What is the total amount Krithika has, if she has the following numbers of notes of ₹ 100, ₹ 20 and ₹ 5? Complete the following table :

No. of ₹ 100 notes	No. of ₹ 20 notes	No. of ₹ 5 notes	Expression and total amount
3	5	6	
			$6 \times 100 + 4 \times 20 + 3 \times 5 = 695$
8	4	z	
x	y	z	

4. Venkatalakshmi owns a flour mill. It takes 10 seconds for the roller mill to start running. Once it is running, each kg of grain takes 8 seconds to grind into powder. Which of the

expressions below describes the time taken to complete grind 'y' kg of grain, assuming the machine is off initially?

(a) $10 + 8 + y$

(b) $(10 + 8) \times y$

(c) $10 \times 8 \times y$

(d) $10 + 8 \times y$

(e) $10 \times y + 8$

5. Write algebraic expressions using letters of your choice.

(a) 5 more than a number

(b) 4 less than a number

(c) 2 less than 13 times a number

(d) 13 less than 2 times a number

6. Describe situations corresponding to the following algebraic expressions:

(a) $8 \times x + 3 \times y$

(b) $15 \times j - 2 \times k$

7. In a calendar month, if any 2×3 grid full of dates is chosen as shown in the picture, write expressions for the dates in the blank cells if the bottom middle cell has date 'w'.

November 2024

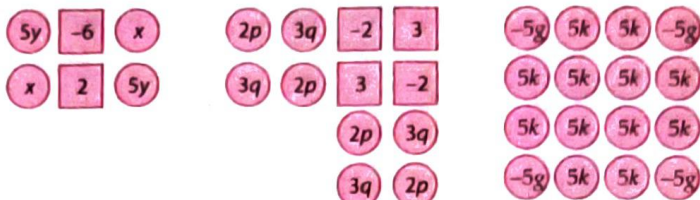
Mon	Tue	Wed	Thu	Fri	Sat	Sun
					1	2
3	4	5	6	7	8	9
10	11	12	13	14	15	16
17	18	19	20	21	22	23
24	25	26	27	28	29	30

$w-1$	w	

Figure it Out

(Page 93 & 94)

1. Add the numbers in each picture below. Write their corresponding expressions and simplify them. Try adding the numbers in each picture in a couple of different ways and see that you get the same thing.



2. Simplify each of the following expressions:

- $p + p + p + p, p + p + p + q, p + q + p - q$
- $p - q + p - q, p + q - p + q$
- $p + q - (p + q), p - q - p - q$
- $2d - d - d - d, 2d - d - d - c$
- $2d - d - (d - c), 2d - (d - d) - c$
- $2d - d - c - c$

Figure it Out

(Page 102-105)

1. One plate of Jowar roti costs ₹ 30 and one plate of Pulao costs ₹ 20. If x plates of Jowar roti and y plates of pulao were ordered in a day, which expression(s) describe the total amount in rupees earned that day?

- $30x + 20y$
- $(30 + 20) \times (x + y)$
- $20x + 30y$
- $(30 + 20) \times x + y$
- $30x - 20y$

2. Pushpita sells two types of flowers on Independence day: champak and marigold. 'p' customers only bought champak, 'q' customers only bought marigold, and 'r' customers bought both. On the same day, she gave away a tiny national flag to every customer. How many flags did she give away that day?

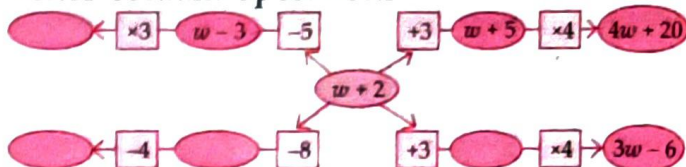
- $p + q + r$
- $p + q + 2r$
- $2 \times (p + q + r)$
- $p + q + r + 2$
- $p + q + r + 1$
- $2 \times (p + q)$

3. A snail is trying to climb along the wall of a deep well. During the day it climbs up 'u' cm and during the night it slowly slips down 'd' cm. This happens for 10 days and 10 nights.

- Write an expression describing how far away the snail is from its starting position.
- What can we say about the snail's movement if $d > u$?

4. Radha is preparing for a cycling race and practices daily. The first week she cycles 5 km every day. Every week she increases the daily distance cycled by 'z' km. How many kilometers would Radha have cycled after 3 weeks?

5. In the following figure, observe how the expression $w + 2$ becomes $4w + 20$ along one path. Fill in the missing blanks on the remaining paths. The ovals contain expressions and the boxes contain operations.



6. A local train from Yahapur to Vahapur stops at three stations at equal distances along the way. The time taken in minutes to travel from one station to the next station is the same and is denoted by t . The train stops for 2 minutes at each of the three stations.

- If $t = 4$, what is the time taken to travel from Yahapur to Vahapur?

- What is the algebraic expression for the time taken to travel from Yahapur to Vahapur? [Hint: Draw a rough diagram to visualise the situation]

7. Simplify the following expressions:

- $3a + 9b - 6 + 8a - 4b - 7a + 16$
- $3(3a - 3b) - 8a - 4b - 16$
- $2(2x - 3) + 8x + 12$
- $8x - (2x - 3) + 12$
- $8h - (5 + 7h) + 9$
- $-23 + 4(6m - 3n) - 8n - 3m - 18$

8. Add the expressions given below:

- $4d - 7c + 9$ and $8c - 11 + 9d$
- $-6f + 19 - 8s$ and $-23 + 13f + 12s$
- $8d - 14c + 9$ and $16c - (11 + 9d)$
- $6f - 20 + 8s$ and $23 - 13f - 12s$
- $13m - 12n$ and $12n - 13m$
- $-26m + 24n$ and $26m - 24n$

9. Subtract the expressions given below:

- $9a - 6b + 14$ from $6a + 9b - 18$
- $-15x + 13 - 9y$ from $7y - 10 + 3x$
- $17g + 9 - 7h$ from $11 - 10g + 3h$
- $9a - 6b + 14$ from $6a - (9b + 18)$
- $10x + 2 + 10y$ from $-3y + 8 - 3x$
- $8g + 4h - 10$ from $7h - 8g + 20$

10. Describe situations corresponding to the following algebraic expressions:

(a) $8x + 3y$

(b) $15x - 2x$

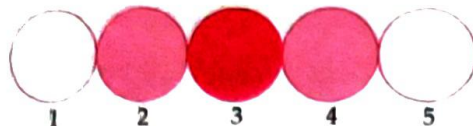
11. Imagine a straight rope. If it is cut once as shown in the picture, we get 2 pieces. If the rope is folded once and then cut as shown, we get 3 pieces. Observe the pattern and find the number of pieces if the rope is folded 10 times and cut. What is the expression for the number of pieces when the rope is folded r times and cut?



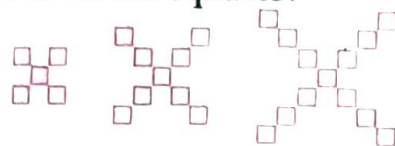
12. Look at the matchstick pattern below. Observe and identify the pattern. How many matchsticks are required to make 10 such squares? How many are required to make w squares?



13. Have you noticed how the colours change in a traffic signal? The sequence of colour changes is shown below. Find the colour at positions 90, 190, and 343. Write expressions to describe the positions for each colour.



14. Observe the pattern below. How many squares will be there in Step 4, Step 10, Step 50? Write a general formula. How would the formula change if we want to count the number of vertices of all the squares?



15. Numbers are written in a particular sequence in this endless 4-column grid.

1	2	3	4
1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

(a) Give expressions to generate all the numbers in a given column (1, 2, 3, 4).

(b) In which row and column will the following numbers appear :

(i) 124 (ii) 147 (iii) 201

(c) What number appears in row r and column c ?

(d) Observe the positions of multiples of 3.

Do you see any pattern in it? List other patterns that you see.

Exercise

Multiple Choice Questions

LEVEL - 1

Formation and Simplification of Expressions

1. If each "T" and each "L" required 2 matchsticks. If a student makes t T's and l L's, then the expression for total matchsticks used is

(a) $t + l$

(c) $t + 2l$

(b) $2t + 2l$

(d) $2(t + l) + 1$

(NCERT Page 82)

2. The score of Ishita in Mathematics is 25 more than the two third of her score in Science. If she scored x marks in Science, determine her score in Mathematics.

(a) $\frac{3}{2}x + 25$

(b) $\frac{2}{3} + 25x$

(c) $\frac{2}{3}x + 25$

(d) $\frac{3}{2} + 25x$ (Page 82)

3. "12 less than 10 times x " is written as

(a) $12 - 10x$

(b) $10x - 12$

(c) $10(x - 12)$

(d) $10x + 12$ (Page 82)

4. The algebraic expression for '4 times x taken away from one-third of y ' is written as

(a) $4x - \frac{y}{3}$ (b) $\frac{y}{3} - 4x$
(c) $4y - 3x$ (d) $4x - 3y$ (Page 82)

5. Which of the following expressions shows distributivity over subtraction?

(a) $4(x - 3) = 4x - 12$ (b) $x + y = y + x$
(c) $x - y = y - x$ (d) $x(y + z) = xy + z$
(Page 85)

6. Which step shows the associative property?

(a) $(x + 2) + 5 = x + (2 + 5)$
(b) $x + 2 = 2 + x$
(c) $x(y + z) = xy + xz$
(d) $x + 0 = x$ (Page 85)

7. Which of the following shows that order does not matter in addition?

(a) $a - b = b - a$ (b) $a + b = b + a$
(c) $ab = ba$ (d) None of these
(Page 85)

8. Which of the following expression is same as $6a + b$?

(a) $6 \times (a + b)$ (b) $(6 \times a) + b$
(c) $6 + ab$ (d) $(6 + a) \times b$
(Page 86)

9. Which of the following is not an example of hidden multiplication?

(a) $4x$ (b) $3y$
(c) $x + 7$ (d) $6b$ (Page 86)

10. The length of a rectangle is $(7x - 3y)$ units and its breadth is $(3x + 4y)$ units. The perimeter of the rectangle is

(a) $(13x + 5y)$ units (b) $(17x - 2y)$ units
(c) $(20x + 2y)$ units (d) $(10x + y)$ units
(Page 87)

11. Subtracting $3xy + 5yz - 7zx$ from $5xy - 2zx + 10xyz$, we get

(a) $2xy - 5yz + 5zx + 10xyz$
(b) $2xy + 5yz - 5zx + 10xyz$
(c) $2xy - 5yz - 5zx - 10xyz$
(d) $2xy + 5yz + 5zx - 10xyz$ (Page 87) **Ev**

12. Mani spent ₹ $(2a + 3b + 5)$ to buy a shirt and ₹ $(7a - 5b + 3)$ to buy a trouser. How much did he spend in all, if $a = 2$, $b = 3$?

(a) ₹ 10 (b) ₹ 20 (c) ₹ 30 (d) ₹ 40
(Page 87) **An**

13. Subtract $(2a - 3b + 4c)$ from the sum of $(a + 3b - 4c)$, $(4a - b + 9c)$ and $(-2b + 3c - a)$.

(a) $3a + 2b + 4c$ (b) $2a - 2b + 4c$
(c) $3a - 4b - 2c$ (d) $2a + 3b + 4c$

(Page 87) **Ev**

14. A pipe is $(7x - 3)$ metres long. A piece of length $(5x - 7)$ metres is cut from it to use. How much pipe (in metres) is left?

(a) $2x + 4$ (b) $-2x + 7$
(c) $3x + 2$ (d) $2x + 1$

(Page 87) **An**

15. The value of expression $3x^2 + x - 10$, when $x = -1$ is

(a) -8 (b) 8
(c) -7 (d) 12 (Page 87) **Ap**

16. Find the value of $5x - 10$ when $x = 3$.

(a) 1 (b) -1
(c) -5 (d) 5 (Page 87)

17. Find the value of expression $\frac{1}{3}x + 5$, at $x = 27$.

(a) 14 (b) 12
(c) 16 (d) 15 (Page 87)

Pick Patterns and Reveal Relationships

18. A machine squares the input and then adds 2. If input is m , then output is

(a) $m^2 + 2$ (b) $m + 2^2$
(c) $m + 2$ (d) $2m + 2$

(Page 95) **An**

19. A gardener plants 3 rose plants in Row 1, 6 in Row 2, 9 in Row 3. Which expression gives number of rose plants in Row r ?

(a) $3 + r$ (b) $3r$
(c) $3r + 1$ (d) $2r + 3$

(Page 96) **An**

20. A pattern has:

Step 1: 5 dots
Step 2: 9 dots
Step 3: 13 dots

What expression gives number of dots in Step n ?

(a) $4n$ (b) $5n$
(c) $4n + 1$ (d) $n + 4$

(Page 96) **An**

21. In any 3×3 square of a calendar, what is the sum of all 9 dates?
 (a) It is always a multiple of 9.
 (b) It is always an even number.
 (c) It is always an odd number.
 (d) It is always a prime numbers.

(Page 97)

22. If 1st November 2024 is Friday, what day will 8th November 2024 be?
 (a) Friday (b) Saturday
 (c) Thursday (d) Sunday (Page 97)

23. What do you observe about diagonal sums in any 2×2 square of a calendar?
 (a) Diagonal sums are different.
 (b) Diagonal sums are always equal.
 (c) Only one diagonal adds up to a prime number.
 (d) None of these

(Page 97)

An

24. Using algebra, what is the expression for the diagonal sum in a 2×2 calendar square with top-left number a ?

- (a) $a + (a + 1)$
 (b) $(a + 8) + (a + 1) = 2a + 9$
 (c) $2a + 7$
 (d) None of these

(Page 97)

U

25. If the centre of the plus-shaped pattern in calendar is a , what is the total of all 5 numbers?

- (a) $a + 4$ (b) $4a$
 (c) $5a$ (d) $a^2 + 4$

(Page 97)

U

26. How many matchsticks are needed to build 5 squares in a line (sharing one side with the next)?

- (a) 20 (b) 16
 (c) 17 (d) 21 (Page 100)

27. Find the number of matchsticks used in n^{th} pattern.



Pattern 1



Pattern 2



Pattern 3

- (a) $5n + 1$ (b) $3n + 1$
 (c) $2n + 1$ (d) $2n + 2$ (Page 100)

28. In pattern 1, there are 6 dots. To form a new pattern, 4 dots are added each time. Find the number of dots in the n^{th} pattern.
 (a) $6n$ (b) $4n + 2$ (c) $2n$ (d) $3n + 3$

LEVEL - 2

29. Tanmay had ₹56. He spent ₹ x on transport and ₹ $2y$ on lunch. He used the remaining amount to buy 3 similar books. What was the cost of each book?

- (a) ₹ $(56 - x + 2y)$ (b) ₹ $(56 - x - 2y)$
 (c) ₹ $\frac{(56 - x + y)}{3}$ (d) ₹ $\frac{(56 - x - 2y)}{3}$

30. Ahmed starts from Delhi at 8 A.M. to Jaipur. If his car is running at the speed of x km/hr and at 1 P.M. he observes that he is 20 km away from Jaipur. Find the distance between Delhi and Jaipur.

(Distance = Speed $\times$ Time)

- (a) $(4x + 20)$ km (b) $(4x + 25)$ km
 (c) $(5x + 25)$ km (d) $(5x + 20)$ km

31. The perimeter of a triangle is $(7x - 15)$ cm. The two sides are $(3x - 3)$ cm and $(2x + 2)$ cm. Find the length of third side.

- (a) $(3x - 13)$ cm (b) $(2x - 14)$ cm
 (c) $(2x - 15)$ cm (d) $(3x - 15)$ cm

32. Somya bought 2 toy cars for ₹ $(4x + 3y)$ each and 1 book for ₹ $(7x - 3y)$. What is the total money (in ₹) spent by her?

- (a) $11x + 6y$ (b) $15x + 3y$
 (c) $3x - 6y$ (d) $3x + 6y$

33. 10 subtracted from three-fourth of x can be expressed as

- (a) $10 - \frac{3x}{4}$ (b) $\frac{3x}{4} - 5$
 (c) $\frac{3x}{4} - 10$ (d) $\frac{4x}{3} - 10$

34. If cost of each pen is ₹ w , find the cost of 40 such pens.

- (a) ₹ $40w$ (b) ₹ $\left(\frac{5w}{8}\right)$
 (c) ₹ $(w + 40)$ (d) ₹ $\left(\frac{w}{40}\right)$

35. 'The sum of twice a number subtracted from 5 and thrice the same number added to 4' can be expressed as

- (a) $(2x + 5) - (4 - 3x)$
 (b) $(2x - 5) + (4 - 3x)$
 (c) $(2x - 5) + (4 + 3x)$
 (d) $(5 - 2x) + (3x + 4)$

U

36. If $x = -3$ and $y = -4$, then the value of expression $24x - 42y + 27$ is

- (a) -69 (b) 123
 (c) -213 (d) 213

Ev

37. The sum of "3 less than 3 times p^2 " and "17 less than $12q^2$ " can be expressed as

- (a) $3p^2 + 12q^2 + 20$ (b) $3p^2 - 3 + 12q^2$
 (c) $3p^2 + 12q^2 - 20$ (d) $p^2 + 12q^2 - 20$

38. A candle burns 3 cm every hour. If its initial height is 20 cm, what is the expression for height of candle after h hours?

- (a) $(20 - 3h)$ cm (b) $(3h - 20)$ cm
 (c) $(20 + 3h)$ cm (d) $(h - 3)$ cm

Ev

39. If an input m passes through a machine that first subtracts 5 and then doubles the result, what is the output expression?

- (a) $2m - 5$ (b) $2(m - 5)$
 (c) $m - 10$ (d) $m - 5 + 2$

Ap

40. Which of the following does not correctly use the distributive property?

- (a) $2(x + 4) = 2x + 8$ (b) $3(y - 5) = 3y - 15$
 (c) $4(z + 1) = 4z + 1$ (d) $5(x + y) = 5x + 5y$

41. Why do we avoid using multiplication " $\times$ " symbol in algebra?

- (a) It's create confusion with addition.
 (b) Letters can look like 'x' and create confusion.
 (c) It's makes the expression longer to write.
 (d) It doesn't work for multiplication.

42. In a calendar, if the top-left number in a 2×2 square is a , what are the other three numbers?

- (a) $a + 1, a + 7, a + 8$
 (b) $a - 1, a - 7, a - 8$
 (c) $a + 2, a + 8, a + 10$
 (d) $a, a + 2, a + 10$

LEVEL - 3 (HOTS)

Competency Focused Questions (CFQs)

43. Find the value of expression

$(x - 6) - (x - 5) + 40 - 17(x - 2)$ at $x = \frac{1}{17}$ after simplifying it.

- (a) 73 (b) -4
 (c) 75 (d) 72

Ev

44. Simplify the given expression and find its value at $x = -44$:

$2(x - 4) - (x + 4) + 54 + (x - 5) - (x - 10)$

- (a) 4 (b) 8
 (c) -8 (d) 3

Ev

45. Read the following statements carefully and select the correct option.

Statement-1 : Difference between the value of expression, $4x + 2(x + y)$ when $x = 2$; $y = 3$ and when $x = 3$; $y = 2$ is 10.

Statement-2 : $4(a + 2b) + 3b^2 + 9ab$ should be subtracted from $2a - 2b + b^2 - 6ab$ to get $-(2a + 10b + 8b^2 + 5ab)$.

- (a) Statement-1 is true but Statement-2 is false.
 (b) Statement-1 is false but Statement-2 is true.
 (c) Both Statement-1 and Statement-2 are true.
 (d) Both Statement-1 and Statement-2 are false.

An

46. The cost of a book is ₹ $(8p + 4q)$, cost of a notebook is ₹ $(5p - 2q)$ and cost of a pen is ₹ $(2p - 2q)$. What is the total cost (in ₹) of all three items ?

- (a) $15q$ (b) $15p + 2q$
 (c) $15p$ (d) $15p - 2q$

An

47. The breadth of a rectangular bedsheet is 5 cm more than half the length of the bed sheet. What is the perimeter of the bed sheet, if the length is x cm?

- (a) $(3x + 12)$ cm (b) $2(x + 5)$ cm
 (c) $(3x + 10)$ cm (d) $(4x + 12)$ cm

48. Ali will be $6y$ years old after 6 years. How old was Ali, $4y$ years ago?

- (a) $(6y - 6)$ years
 (b) $(10y - 6)$ years
 (c) $(2y - 6)$ years
 (d) $(2y + 6)$ years

U

49. A ladder is built with rungs. Each rung adds 3 screws. The first rung takes 5 screws. Which algebraic expression will represent total screws for n rungs?

- (a) $5n$ (b) $3n + 5$
(c) $5 + 3(n - 1)$ (d) $3n - 1$

50. A student made letter 'H' using 6 matchsticks. He student places m H's sharing both vertical sides. How many matchsticks are needed for this?

- (a) $6m$ (b) $6 + 4(m - 1)$
(c) $6 + 5(m - 1)$ (d) $3m + 3$



Match the Following

In this section, each question has two matching lists. Choices for the correct combination of elements from List-I and List-II are given as options (a), (b), (c) and (d) out of which one is correct.

1. Match the following.

- | List-I | List-II |
|---------------------------------|---------------------------|
| (P) $a + b = b + a$ | (1) Associative Property |
| (Q) $3(x - y) = 3x - 3y$ | (2) Distributive Property |
| (R) $(x + 2) + 4 = x + (2 + 4)$ | (3) Commutative Property |
| (a) P-2, Q-1, R-3 | (b) P-3, Q-1, R-2 |
| (c) P-3, Q-2, R-1 | (d) P-1, Q-2, R-3 |

2. Match the pattern description as given in List-I with algebraic expression for n patterns as given in List-II.

- | List-I | List-II |
|--|--------------------|
| (P) One zigzag shape needs 5 matchsticks. Each extra repeat adds 3 more match sticks | (1) $6 + 4(n - 1)$ |
| (Q) Letter 'H' is made using 6 sticks. Each new 'H' shares only 2 matchsticks. | (2) $4n$ |
| (R) Letter 'F' is made using 4 matchsticks. | (3) $5 + 3(n - 1)$ |
| (a) P-1, Q-3, R-2 | (b) P-3, Q-1, R-2 |
| (c) P-3, Q-2, R-1 | (d) P-1, Q-2, R-3 |

3. Match the input and output with the given algebraic expressions.

- | List-I | List-II |
|----------------------------|------------------------|
| (P) Input = 5, Output = 32 | (1) $7x + 2$ |
| (Q) Input = 4, Output = 30 | (2) $8x - 4$ |
| (R) Input = 3, Output = 20 | (3) $7x - 3$ |
| (S) Input = 2, Output = 0 | (4) $9x - 18$ |
| (a) P-1, Q-3, R-4, S-2 | (b) P-3, Q-1, R-2, S-4 |
| (c) P-3, Q-2, R-4, S-1 | (d) P-1, Q-2, R-3, S-4 |

4. Match the value of given expressions.

- | List-I | List-II |
|---|------------------------|
| (P) $(x^2 + 5)(x^3 + 3) + 5$ at $x = 2$ | (1) 0 |
| (Q) $2x^3 + 2x^2 - 4x - 4$ at $x = -2$ | (2) 104 |
| (R) $\left(\frac{-10}{3}xy^3\right) + \left(\frac{6}{5}x^3y\right)$ at $x = 5, y = 3$ | (3) 13 |
| (S) $x^3 - x^2 - x - 2$ at $x = 3$ | (4) -4 |
| (a) P-3, Q-2, R-4, S-1 | (b) P-2, Q-3, R-1, S-4 |
| (c) P-2, Q-4, R-1, S-3 | (d) P-3, Q-1, R-4, S-2 |

Assertion & Reason Type

Directions : In each of the following questions, a statement of Assertion is given followed by a corresponding statement of Reason just below it. Of the statements, mark the correct answer as

- (a) If both assertion and reason are true and reason is the correct explanation of assertion.
(b) If both assertion and reason are true but reason is not the correct explanation of assertion.
(c) If assertion is true but reason is false.
(d) If assertion is false but reason is true.

1. Assertion : The value of $19x - 5x^2$ for $x = -2$ is -18 .

Reason : The value of the expression depends on the value of the variable from which the expression is formed.

2. Assertion : If $p = 4$, $q = 7$, then the value of pq is 21.

Reason : The expression pq indicates multiplication of p and q .

3. Assertion : $5(x + y) = 5x + 5y$ shows the distributive property.

Reason : Distributive property says: $a(b + c) = ab + ac$.

4. Assertion : Associative property changes the order of numbers.

Reason : $(a + b) + c = a + (b + c)$ shows associative property.

5. Assertion : To make n squares in a straight row (sharing a side), the number of matchsticks needed is $4n + 1$.

Reason : Each adjacent square requires 3 matchsticks.

6. Assertion : The expression $2a + 3b$ is simplified form of $2 \times a + 3 \times b$.

Reason : Algebra avoids using $\times$ sign when multiplying numbers with letters.

7. Assertion : A rope is folded 4 times and cut once. It gives 5 pieces.

Reason : The number of pieces = Number of folds + 1.

8. Assertion : In a machine, for input 6, output is 10 and for input 7, output is 12. So, the expression on which machine works is $x + 4$.

Reason : When $x = 6$, then $x + 4 = 6 + 4 = 10$.

9. Assertion : A rectangular floor is made using square tiles of area of y sq. cm. If there are r rows and each row contains $2r + 1$ tiles, then the algebraic expression for the total area (in sq. cm) of the floor is $y^2 \times r \times (2r + 1)$.

Reason : By using letters to represent quantities, we cannot write general formulas that help in any situation.

10. Assertion : The expression $3x + 2y$ and $2x + 3y$ are equal.

Reason : Algebraic expressions must maintain the term order to satisfy the commutative property.



Comprehension Type

PASSAGE-I : While playing with a calendar, Arjun noticed something interesting: Whenever he picked any 2×2 square grid of dates, the diagonal sums are equal. For example:

12	13
19	20

$$12 + 20 = 32$$

$$13 + 19 = 32$$

He used algebra to explain the pattern. If the bottom right number is a , then 2×2 grid is of the form

$a - 8$	$a - 7$
$a - 1$	a

1. What is the diagonal sum for a 2×2 square with top-left number 18?

(a) 36 (b) 44 (c) 38 (d) 28

2. The sum of the dates in one diagonal of a 2×2 Calendar square is x . What is the sum of the other diagonal?

(a) x (b) $x + 1$ (c) $x + 2$ (d) $x - 1$

3. What is the general expression for diagonal sum using bottom-left number a ?

(a) $2a$ (b) $a - 8$
(c) $2a - 6$ (d) $2a + 8$

PASSAGE-II : A machine is based on an algebraic expression, when we input two values at a time, then based on that expression, machine gives some output. The following table shown some input and corresponding output.

Input (x)	Input (y)	Output
5	4	19
6	8	29
3	6	19
7	7	29

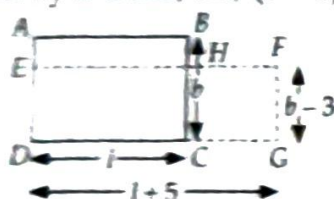
1. Find the algebraic expression on which machine is working.

(a) $(x + y) + 10$ (b) $2x + 3y - 3$
(c) $2(x + y) + 1$ (d) $2(x + y) - 1$

2. If $x = 10$ and $y = 11$, then output will be
(a) 43 (b) 31 (c) 19 (d) 29

3. Which of the following can not be the output for any value of x any y ?
(a) 2 (b) 6
(c) 0 (d) All of these

PASSAGE-III : We have a rectangle of length l and breadth b . The length of the rectangle is increased by 5 units i.e., $(l + 5)$ units and breadth is decreased by 3 units, i.e., $(b - 3)$ units.



What is the perimeter of rectangle DEHC?

(a) $(3l + 2b + 6)$ units (b) $(2l + 2b - 6)$ units
(c) $(3l - 2b - 6)$ units (d) $(2l - 3b - 15)$ units

If length of rectangle DEFG is 2 times its breadth, then find the perimeter of the rectangle DEFG.

(a) $(6b + 16)$ units (b) $(2b + 18)$ units
(c) $(6b - 18)$ units (d) $(6l + 16)$ units

3. What is the perimeter of rectangle DEFG?

(a) $(2l + 2b + 4)$ units (b) $(2l + 2b)$ units
(c) $(2l + 4b - 2)$ units (d) $(2b + 4l + 2)$ units

Subjective Problems

Very Short Answer Type

1. Look at the patterns and fill up the missing boxes. State the rule which connects the quantities in the different columns. Find the generalised statement for n in each case.

a	15	14	13	12	11	10	n
b	10	9	8	7			

2. If the cost of painting 1 sq. m of wall is ₹60, what is the cost for a wall of area l b sq. m, where l is length of wall and b is breadth of wall?

3. A troop of soldiers is standing in rows. In each row, there are 27 soldiers. Find the rule that can be used to calculate the total number of soldiers in x rows.

4. A banner shows letters made from matchsticks. Each "K" uses 4 matchsticks, and each "M" uses 6 matchsticks. If a student places x Ks and y Ms, write the expression for total matchsticks.

5. "25 taken away from the product of x and y " write in the form of algebraic expression.

6. The length of a rectangular wall is l cm, the breadth is $2b$ cm. Find the area of the wall in sq. cm.

7. If $a = -2$, $b = -3$, $c = 2$, then find the value of $\frac{ab}{c} + 9$?

8. Write the statement for given expression $\frac{P}{5} + 1$.

Short Answer Type

1. The middle row of 3×3 calendar square has values $a - 1$, a , $a + 1$. Write the values for the top and bottom rows using calendar pattern logic.

2. A week in a calendar is marked: 12, 13, 14, 15, 16, 17, 18. Find the sum of all dates. Can you write an expression, if the first date is a ?

3. A ladder is made with 4 vertical matchsticks and r horizontal rungs, each using 3 matchsticks. Write an expression for the total matchsticks required to build such a ladder.

4. The length and breadth of a rectangle are $(6x - 2)$ units and $(5x + 2)$ units respectively. Find its perimeter.

5. Simplify : $(15x - 4y) + (15x + 4y + 3) - 45x + 12y$

6. If $q = 5$ and $p = -3$, then find the value of the following expressions.

(i) $q + 3p - 10$ (ii) $8q + 9p - 17$

7. If 1st May is Monday, then what is the day on 18th May?

8. A shopkeeper earns ₹ $(3pq - 2p + q - 2)$ on Monday and loses ₹ $(3pq + 6q - 2)$ on Tuesday. What is his net earning? Does shopkeeper is having overall profit or loss?

Long Answer Type

- Simplify : $(2x + 3y) - (3x + 4y) + (7x + 3y) + (x + 2y)$ and find value for $x = 2$, $y = 1$.
- How can we write 7 times a number decreased by 15 in algebraic expression? What is thrice this expression?
- Find the value of $10m - 16n + 2p + 5$, if $m = 6$, $n = -4$ and $p = 3$.
- In a calendar, 2×3 grid of dates is chosen such that the bottom-middle cell has the date "w."
 - Express the other five cells in terms of w.
 - Write an expression for the sum of all 6 dates.
 - Is this sum always divisible by 6? Why or why not?
- Each letter in the word "HERO" is made by using matchsticks: H = 6, E = 5, R = 6, O = 6 matchsticks. When these letters are repeated n times to create a banner:
 - Write an expression for total matchsticks used.
 - If every repeated "HERO" share 2 match sticks with the previous one, modify your expression.
 - Calculate value of expression obtained in (ii) for $n = 8$.

Integer/Numerical Value Type

- In a 2×2 calendar square, the top-left number is 15. What is the total sum of the two diagonals?
- Evaluate : $3(x + 2y) - 2(x - y)$ for $x = 4$, $y = 2$.
- If a 3×3 calendar square starts with number 10 at top-left, what is the sum of all the 9 values?
- Simplify and evaluate: $3(x + 2) + 2(x - 1)$ for $x = 4$.

- In staircase pattern, Step 1: 2 tiles, Step 2: 5 tiles and Step 3: 8 tiles. What is the number of tiles at Step 6?
- A rectangular table has length l and width w . If 3 such tables are joined end-to-end (length-wise), the new table has extra supports only at ends. Write expression for total perimeter and calculate it for $l = 4$, $w = 2$.
- A maze is made by connecting hexagonal tiles. Each hexagon uses 6 matchsticks. When connected in a line, each new hexagon shares 2 matchsticks with the previous. Write an expression for the number of matchsticks needed for 10 hexagons.
- A rope is 80 cm long. You cut x pieces from it, each of length 6 cm. Write an expression for the remaining length of rope and find its length when $x = 9$.



Case Based Questions

Case I : Ms. Isha is designing a large mural on a school wall using square tiles. Each row in the mural has different pattern in the number of tiles used. Here's how it grows :

Row Number (r)	Number of Tiles (T)
1	5
2	9
3	13
4	17
5	21

Based on above information, answer the following questions.

- How many tiles are increasing in each row as compared to the previous row?
 - 3
 - 4
 - 5
 - Not a fixed number
- Which expression gives the number of tiles in row r ?
 - $4r$
 - $4r - 1$
 - $5r - 1$
 - $4r + 1$
- What will be the number of tiles in the 10th row?
 - 41
 - 44
 - 40
 - 45

4. Ms. Isha combines first and third rows. What expression represents total tiles?
 (a) $(4 \times 1 + 1) + (4 \times 3 + 1)$
 (b) $5 + 15$
 (c) $4(1 + 3) + 1$
 (d) $4 \times 4 + 1$

5. If a student wrongly writes the pattern as $5 + 2(r - 1)$, what tile count will they get for row 4?
 (a) 11 (b) 13 (c) 15 (d) 17

Case II : Riya and Aarav are working on a math puzzle involving monthly calendar grids. They explore the secret of picking 4×4 squares (a block of 16 dates) from any monthly calendar grids.

4	5	6	7
11	12	13	14
18	19	20	21
25	26	27	28

They try to observe many things by assigning algebraic expression to the dates.

Based on the given information, answer the following questions.

- Sum of elements of both the diagonals of 4×4 grid is
 (a) 68 (b) 66
 (c) Both the diagonals have equal sum.
 (d) Both the diagonals have different sum.
- If top left corner is a , then the bottom right corner is
 (a) $a + 28$ (b) $a + 26$
 (c) $a + 24$ (d) $a + 21$
- If top left corner is represented by a , then the expression representing sum of diagonal is

- (a) $4a + 40$ (b) $4a + 48$
 (c) $4a + 52$ (d) None of these

4. If a 3×3 grid is selected from the given 4×4 grid have centre value 19, then what will be the bottom right value in the grid?
 (a) 27 (b) 28
 (c) 25 (d) 26
5. How many different 2×2 grids can be made from the given 4×4 grid?
 (a) 6 (b) 7
 (c) 8 (d) 9

Case III : In a fruit store, different boxes designated for three different fruits. There are n boxes of mangoes, each containing 15 mangoes. Apples are placed in 4 boxes with x number of apples in each box. Additionally, there are 6 boxes of guavas, each containing 20 guavas. Based on the above information, answer the following questions.

- Total number of mangoes in n boxes is
 (a) $15 + n$ (b) $15n$
 (c) $15 - n$ (d) None of these
- Total number of apples in 4 boxes is
 (a) $4x$ (b) $4 + x$ (c) $4 - x$ (d) $4 \div x$
- The algebraic expression representing the total number of fruits in the store is equal to
 (a) $15n + 4x + 120$ (b) $15x + 4n + 100$
 (c) $15x + 4n + 120$ (d) $15n + 4x + 100$
- Find the total number of mangoes in the store if $n = 10$.
 (a) 100 (b) 50 (c) 150 (d) 200
- Find the total number of fruits in the store if $n = 10$ and $x = 20$.
 (a) 100 (b) 250 (c) 150 (d) 350